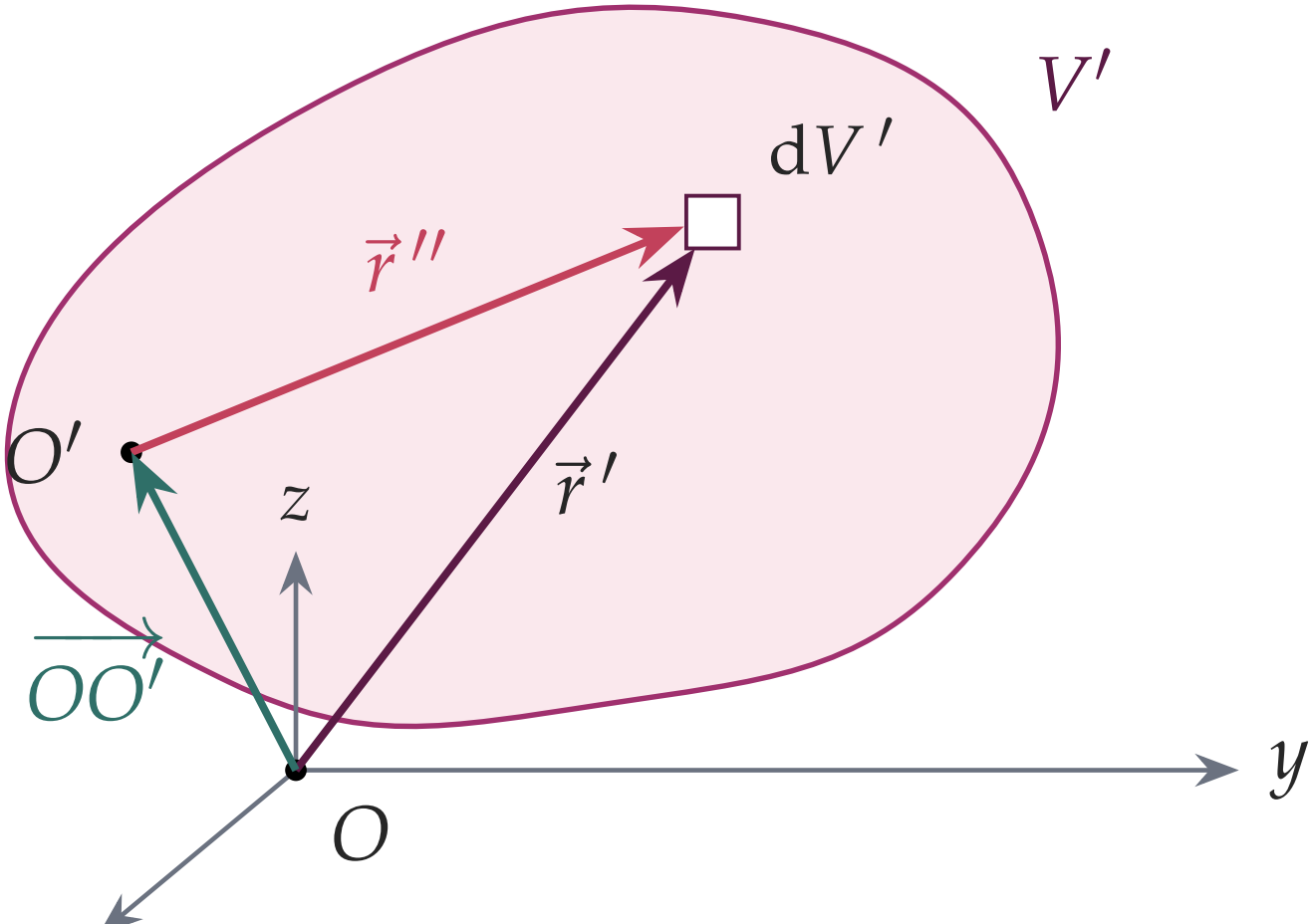

$$\vec{r}\,'' = \vec{r}\,' - \overrightarrow{OO'}$$

$$\vec{p}_{O'} = \vec{p}_O - Q\,\overrightarrow{OO'}$$